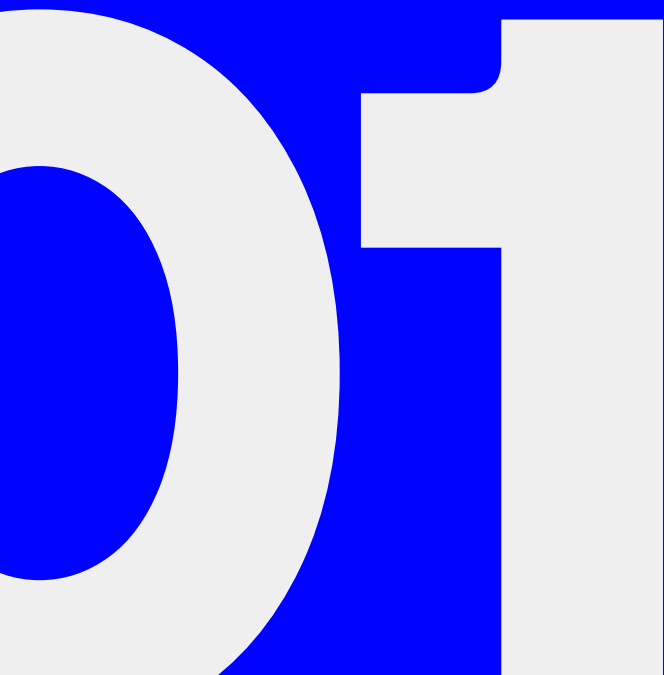


OpenSec-Sentinel: AI-Driven Intrusion Prevention & SOAR

Presented by Rejeesh Koshy



Origin of the idea

The Problem with Traditional Blue Team Operations

Static Rules Fail: Traditional tools (like Fail2Ban) rely on static thresholds. Attackers bypass these using "low-and-slow" distributed attacks

Alert Fatigue: Human analysts are overwhelmed by raw logs, leading to missed zero-day threats.

Slow MTTR: Mean Time to Respond (MTTR) is dictated by human speed. Manual firewall updates are too slow to stop automated botnets.

The OpenSec-Sentinel Architecture

A Closed-Loop Security Pipeline

01.

The Brain: Python-based AI monitoring network traffic in real-time.

02.

The Nerve Center: Dockerized Wazuh SIEM ingesting AI telemetry via native host syslog.

03.

The Muscle: Wazuh Active Response automatically neutralizing threats at the network layer.

Threat Detection via Machine Learning

Model Selection: Isolation Forest. Excels at explicitly isolating statistical anomalies rather than profiling "normal" traffic. Extremely low-latency

Training Methodology: Synthetically generated data mapping

Mathematical Baseline: Uses a Poisson distribution to accurately simulate the random, independent arrival rate of real-world network packets.

Automated Threat Detection

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Closed-Loop Remediation



SOAR

Security Orchestration, Automation, and Response

Container Bypass: AI telemetry is piped directly into the host's native syslog, bypassing Docker isolation boundaries for seamless agent forwarding.

Custom Rule Parsing: Wazuh Manager evaluates Rule 100001 triggered by the AI's BLOCK command

Automated Remediation: Instant execution of the active-response/bin/firewall-drop script, modifying iptables to block the attacker in milliseconds.

04

■ 01

Baseline Traffic
Monitoring

■ 03

Layer-7 SSH Brute-
Force Attack

■ 02

AI Anomaly Detection

■ 04

Automated Firewall
Drop

System Validation

Closed-Loop Remediation

```
sysadmin@sentinel-core: ~/wazuh-docker/single-node
sysadmin@sentinel-core: ~/wazuh-docker/single-node$ sudo docker exec single-node-wazuh.manager-1 tail -f /var/ossec/logs/alerts/alerts.json | grep -A5 "100001"
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```

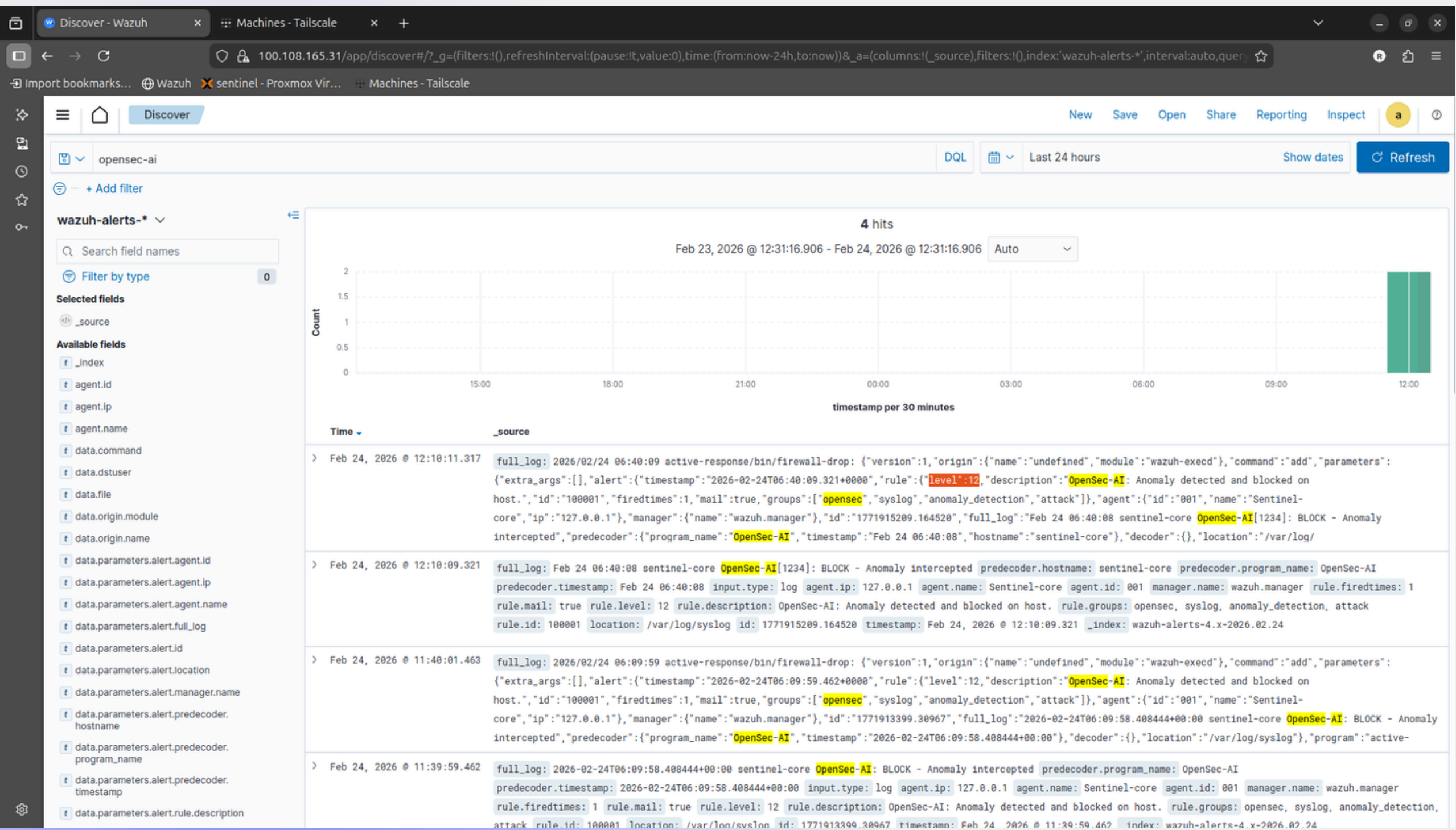

Automated Threat Detection

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Project

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Closed-Loop Remediation



```
sysadmin@sentinel-core: ~
sysadmin@sentinel-core: ~/wazu...
sysadmin@sentinel-core: ~
rejeesh.koshy@rejeesh-koshy-Thi...

_samples for estimation.
warn(
Logged: MONITOR (Traffic: 2)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 3)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 5)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 9)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 8)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 10)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 8)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 8)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
samples (100) is greater than the total number of samples (3). max_samples will be set to n
_samples for estimation.
warn(
Logged: MONITOR (Traffic: 8)
/usr/local/lib/python3.12/dist-packages/sklearn/ensemble/_iforest.py:344: UserWarning: max_
```




Final reflections and future steps

Achieves a near-zero MTTR without human intervention.

Replaces easily evaded signature-based rules with behavioral statistical analysis.